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RULix: A Foundation Model for Zero-Shot Remaining Useful Life Prediction

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Abstract

Data-driven remaining useful life (RUL) prediction can reduce unplanned downtime in manufacturing, yet it requires run-to-failure histories that production assets rarely provide. Machines are maintained before failure and fleets are small, so labeled degradation data remain scarce at any single site. Recent studies therefore reuse general-purpose foundation models for prognostics, while pretraining on degradation processes remains largely unexplored. This paper presents a prognostic model pretrained exclusively on synthetic run-to-failure data. In the underlying prior, a latent health index degrades as a stochastic differential equation with load-modulated drift and fails at first passage. Sensor signals are emitted through randomly sampled temporal causal graphs, and units are right-censored with capped RUL labels. A causal extended long short-term memory (xLSTM) encoder is pretrained to estimate health state, degradation dynamics, and RUL from a sliding sensor window. At inference, RUL is predicted directly or by simulating the learned dynamics to first passage. The pretrained model is evaluated zero-shot on the C-MAPSS and N-CMAPSS turbofan benchmarks. The approach is a step toward prognostic foundation models for assets without recorded failure histories.

Keywords: Predictive maintenance; Remaining useful life; Foundation models; Synthetic data; xLSTM

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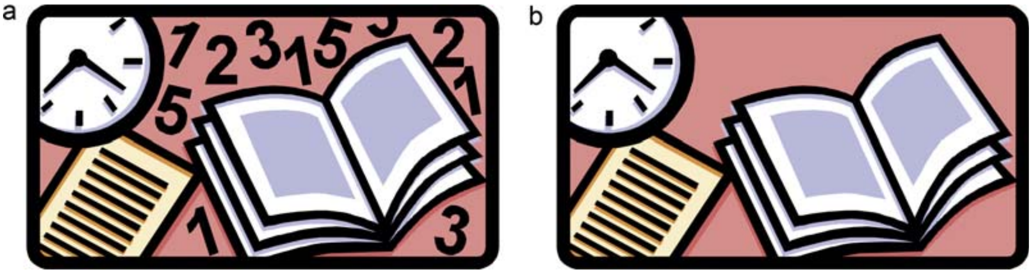

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References

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